# Pre-A Round-1 Inventory and Q3LOCK Weighted-Energy Common-Alpha Advance

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

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## 1 Purpose and scope

This checkpoint resumes T-054 after `EXP-000790`. It advances two independent bottlenecks while retaining every no-overclaim boundary.

First, the bounded Round-1 T-053 evidence inventory and common categorical admission contract are frozen. The current M1, M2, and M5 versions contain no admitted microscopic survivor; M0 remains only the established effective baseline. This is non-selection, not universal falsification. The visible helium exponent was public before the new M2 stiffness map was written, so its mismatch is retrospective and receives no validation credit. The parent Round-1 gate remains open.

Second, the exact fixed-lattice ST8/Q3LOCK Hamiltonians now have one common local polynomial derivation and a source-uniform first weighted-local-energy cone. Two tempting unweighted cutoff routes fail quantitatively. Higher weighted-energy moments and a thermodynamic Cauchy theorem remain necessary before one may claim a common real-time automorphism or identify both phasewise systems as KMS states of that same dynamics.

No physical functional, vacuum, continuum, empty-space ordering, GNS gap, light, mass, time, gravity, C6, CP1, Sector-A or Pre-A completion follows.

## 2 Round-1 evidence and admission firewall

Every inherited Round-1 source item has exactly one role. Propagation data constrain speeds and dispersion, not the microscopic number of zero modes, gauge constraints, quotients, or polarizations. Superconducting flux is a mechanism calibration and is not direct vacuum-winding evidence. The helium exponent is a visible retrospective diagnostic rather than a blind holdout.

The missing or deferred families remain explicit: direct vacuum topology, microscopic physical mode count, particle spectrum and charge, universal gravitational response, cosmological cooling, and a normalized physical-empty preparation branch. Thus evidence is not duplicated under different names.

The only categorical outcomes are

$$\{\mathrm{PASS}, \mathrm{FAIL}, \mathrm{NOT\_TESTED}, \mathrm{NOT\_ADMITTED}, \mathrm{INCOMPARABLE}\}.$$

Here `FAIL` is scoped falsification, `NOT_ADMITTED` is an absent required law or map, `NOT_TESTED` is an unexecuted declared test, and `INCOMPARABLE` means no common comparison contract exists. An eligible microscopic candidate survives only if every hard row D00–D09 passes. The survivor set is computed from the matrix rather than copied from the manifest.

D00 is intentionally narrower than full admission: it passes when all nine canonical field labels and every absence or partial status are explicit. It does not certify semantic completeness or the still-open per-parameter common input ledger.

Candidate-internal energies may be compared only under one unchanged law, reference, volume, boundary, regulator, normalization and additive scalar. Raw energies belonging to different Hamiltonians are not ranked.

## 3 Exact Round-1 quadratic checks

For M2 the registered critical kernel is

$$K(k) = r + c\sum_{i=1}^{3}(q^2 - k_i^2)^2.$$

At $k_\star \in \{+q, -q\}^3$, direct expansion gives

$$D_k^2 K(k_\star) = 8cq^2 I_3, \qquad \omega^2 = \frac{4cq^2}{\chi}|p|^2 + O(|p|^3). \tag{3.1}$$

The separately inserted inertial lane therefore has an exact isotropic tree cone. It does not pass the physical speed discriminator: the all-mode observable map, protection mechanism, and candidate-neutral comparison are absent.

M1's registered evolution is first-order $L^2$ gradient flow and declares no canonical momentum or conservative real-time law. Bare M5 has a continuous isotropic-shell spatial Hessian of rank one. Adding inertia, isolated nodes, a compact target, a gauge quotient, or a new observable map defines a new candidate version rather than repairing the scored version silently.

## 4 Why the visible stiffness mismatch is not validation

The retrospective analytic map assumes

$$f(A,t) = \frac{at}{2}A^2 + \frac{u_{\mathrm{eff}}}{4}A^4, \qquad \rho_s = Z(t)A^2, \tag{4.1}$$

where $a, u_{\mathrm{eff}}, Z(0) > 0$ and all coefficients are analytic. Minimizing for $t < 0$ gives

$$A^2 = -\frac{at}{u_{\mathrm{eff}}}, \qquad \rho_s \propto |t|,$$

so the exact tree exponent is one. It lies outside the visible interval $[0.671, 0.673]$. However, that target predates this observable map. Parameter reuse is not evidenced, but map-selection leakage cannot be excluded. The comparison is therefore recorded as a retrospective mechanism conflict with zero validation credit; M2's D07 cell is not scored as a prospective failure.

Closing the parent gate requires a future/blind target or genuinely new relation, together with its microscopic observable map, frozen in an earlier immutable commit before its result is inspected.

## 5 Exact Q3LOCK model and common local derivation

At each spatial site let $q_x, p_x \in \mathbb{R}^8$. The onsite quartic is

$$W_4(q) = \frac{g}{4}\sum_e q_e^4 + \frac{\lambda}{4}\sum_{e\sim f \text{ in } Q_3}(q_e - q_f)^2(q_e^2 + q_f^2), \tag{5.1}$$

with $g > 0$, $\lambda \geq 0$, $c > 0$, and $\chi > 0$. For $u = 8^{-1/2}(1, \ldots, 1)$,

$$H_\Lambda(h) = \sum_{x\in\Lambda}\left[\frac{|p_x|^2}{2\chi} + \frac{r}{2}|q_x|^2 + W_4(q_x) - hu\cdot q_x\right] + \frac{c}{2}\sum_{\langle xy\rangle}|q_x - q_y|^2. \tag{5.2}$$

On the finite-support polynomial CCR star-algebra, define

$$\delta_h(A) = \frac{i}{\hbar}[H_\Lambda(h), A]. \tag{5.3}$$

If $\Lambda$ contains the one-neighbourhood of the support of $A$, the right side is independent of the containing volume. On generators,

$$\delta_h(q_x) = \frac{p_x}{\chi}, \qquad \delta_h(p_x) = -rq_x - \nabla W_4(q_x) - c\sum_{y\sim x}(q_x - q_y) + hu. \tag{5.4}$$

Thus $\delta_0$ is fixed before selecting either sign phase. This is a common local derivation, not yet a thermodynamic-limit C-star automorphism.

# 6 Source-uniform coercivity

The Q3 locking term is nonnegative and

$$W_4(q) \geq \frac{g}{4}\sum_e q_e^4 \geq \frac{g}{32}|q|^4. \tag{6.1}$$

Fix $0 < \gamma < g/32$, put $a_\gamma = g/32 - \gamma$, and define

$$C_\gamma(r, h_0) = \max_{\rho\geq 0}\left[h_0\rho - \frac{r}{2}\rho^2 - a_\gamma\rho^4\right]. \tag{6.2}$$

Since $|u\cdot q| \leq |q|$, for every $|h| \leq h_0$,

$$K_{\Lambda,h} := H_\Lambda(h) + C_\gamma|\Lambda| \geq \sum_x\left[\frac{|p_x|^2}{2\chi} + \gamma|q_x|^4\right] + \frac{c}{2}\sum_{\langle xy\rangle}|q_x - q_y|^2. \tag{6.3}$$

For every $\epsilon > 0$, exact scalar Young optimization gives

$$\sum_{x\in X}|q_x|^2 \leq \epsilon K_{\Lambda,h} + \frac{|X|}{4\epsilon\gamma}, \tag{6.4}$$

$$\pm c q_x\cdot q_y \leq \epsilon K_{\Lambda,h} + \frac{c^2}{8\epsilon\gamma}. \tag{6.5}$$

These constants and the shift are uniform in volume and the compact source interval.

# 7 Weighted first-local-energy theorem

Give half of each bond energy to each endpoint:

$$e_x = \frac{|p_x|^2}{2\chi} + \frac{r}{2}|q_x|^2 + W_4(q_x) - hu\cdot q_x + C_\gamma + \frac{c}{4}\sum_{y\sim x}|q_x - q_y|^2. \tag{7.1}$$

Then $e_x \geq 0$ as a quadratic form and $\sum_x e_x = K_{\Lambda,h}$. On the common Schwartz/form core, onsite forces cancel and

$$\frac{de_x}{dt} = -\sum_{y\sim x} J_{xy}, \qquad J_{xy} = \frac{c}{4\chi}\{p_x + p_y, q_x - q_y\}, \quad J_{yx} = -J_{xy}. \tag{7.2}$$

The total bond momentum commutes with the relative coordinate. The joint kinetic–bond quadratic form completes a square and yields the sharp form bound

$$\pm J_{xy} \le \sqrt{\frac{c}{2\chi}}(e_x + e_y). \tag{7.3}$$

For $E_f = \sum_x f_x e_x$, suppose the positive weights obey $e^{-\mu} \le f_x/f_y \le e^{\mu}$ on adjacent sites. The exact continuity equation, (7.3), and degree six of $\mathbb{Z}^3$ give

$$\pm\delta_h(E_f) \le 6\sqrt{\frac{c}{2\chi}}(e^{\mu} - 1)E_f. \tag{7.4}$$

Finite-volume form-domain Gronwall therefore proves

$$\tau_t^{\Lambda}(E_f) \le \exp\left[6\sqrt{\frac{c}{2\chi}}(e^{\mu} - 1)|t|\right] E_f. \tag{7.5}$$

This is phase-, source-, and volume-uniform at the first energy moment. It does not control all nested cubic-force commutators or make the volume net of automorphisms Cauchy.

# 8 Two exact failed shortcuts

Suppose a Fourier–Stieltjes cutoff satisfies

$$V_R(q) = \int e^{ik\cdot q}\,\mu_R(dk), \qquad \kappa_R = \int |k|^2 |\mu_R|(dk), \tag{8.1}$$

and agrees with $W_4$ on $|q| \le R$. Then

$$\sup_{q,|v|=1} |v^T D^2 V_R(q) v| \le \kappa_R. \tag{8.2}$$

Along a Q3 coordinate ray, exactly three edges meet the occupied vertex:

$$W_4(te) = \frac{g + 3\lambda}{4} t^4, \qquad \kappa_R \ge 3(g + 3\lambda)R^2. \tag{8.3}$$

Thus the speed furnished by the audited global-second-moment bounded-Weyl theorem diverges at least quadratically. This refutes that uniform import, not exact common dynamics.

For a basic resolvent $R_z = (a\cdot P + b\cdot Q - z)^{-1}$, $a \ne 0$ and $\operatorname{Im} z \ne 0$,

$$[W_4(Q), R_z] = -i\hbar R_z (D_a W_4)(Q) R_z. \tag{8.4}$$

Let $L = a\cdot P + b\cdot Q$. Choose normalized $\xi \in C_c^{\infty}(\mathbb{R}^8)$, supported in $B_{r_0}$, and a Weyl unitary $U_s$ with

$$U_s^* Q U_s = Q + sa, \quad U_s^* P U_s = P - sb, \quad U_s^* L U_s = L. \tag{8.5}$$

For $\psi_s = U_s(L - z)\xi$ and $\phi_s = U_s(L - \bar{z})\xi$, both input norms are independent of $s$, and resolvent cancellation gives

$$\langle \phi_s, R_z (D_a W_4)(Q) R_z \psi_s\rangle = 4W_4(a)s^3 + O(s^2), \qquad W_4(a) > 0. \tag{8.6}$$

This fixed-denominator matrix-element lower bound proves that the exact sandwich has no bounded extension. If a smooth cutoff $V_R$ equals $W_4$ on $B_R$, the same identity holds while $0 \le s \le (R - r_0)/|a|$. Taking $s = (R - r_0)/(2|a|)$ yields

$$\|R_z(D_aV_R)(Q)R_z\| \geq c_{\xi,z,a}R^3 \tag{8.7}$$

for all large $R$. This rules out the ordinary unweighted basic-resolvent generator estimate, not an energy-damped core or a different invariance proof.

# 9 Remaining common-alpha gate

On one predeclared energy-damped local resolvent/regular core, prove enough higher weighted-energy graph norms to control the cubic force and an estimate of the form

$$\|(1+E_{\mu,X})^{-s}[\tau_t^{\Lambda'}(A)-\tau_t^{\Lambda}(A)](1+E_{\mu,X})^{-s}\| \leq C_{A,T}e^{-\mu(d(X,\Lambda^c)-vT)}. \tag{9.1}$$

The constants must be uniform in the larger volume, compact source interval, exhaustion and auxiliary cutoff. Only after multiplicativity, invertibility, time continuity and exhaustion independence follow may the two phasewise systems be tested as KMS states of one common $\alpha$.

The minimal falsifier is one fixed local observable and compact time interval for which every finite energy weight fails to make the volume net Cauchy.

# 10 Verification and prior-art boundary

Both result families are checked by a symbolic primary implementation, a non-importing exact-arithmetic implementation, and an integrated freshness and scope audit. The independent routes reconstruct the Q3 graph, quadratic coefficients, survivor set, current coefficient, sharp energy constant, cutoff growth and all negative scope flags from upstream data.

The nearest quantum dynamics theorems use bounded Weyl-integral perturbations, bounded $C_0$ interactions, Schwartz potentials, or subquadratic forces and do not directly close the exact quartic-plus-harmonic model. Classical quartic-crystal local-energy propagation supports the route only; it is not imported as a quantum automorphism theorem.

The two registered failure IDs are `NG-2026-08-10-PRE-A-ST8-Q3LOCK-FOURIER-SECOND-MOMENT-UNIFORM-NORM-LR-CUTOFF` and `NG-2026-08-10-PRE-A-ST8-Q3LOCK-BASIC-RESOLVENT-CUBIC-FORCE-UNWEIGHTED-CORE`.

The exact next common-dynamics subgate is higher weighted-energy moments plus thermodynamic Cauchy closure. The exact next Round-1 requirement is a truly prospective validation relation and microscopic observable map. Pre-A remains open.

# 11 Devil's-advocate audit

1. **Objection: the visible exponent is a preregistered validation. UPHELD as false.** The target predates the new observable map; the final package removes validation credit and keeps the parent gate open.

2. **Objection: no current survivor falsifies all microscopic theories. UPHELD as false.** Only the frozen M1/M2/M5 versions are classified; repairs are new candidates.

3. **Objection: the M2 internal cone already passes the physical speed test. UPHELD as false.** Its inertial law is inserted and the physical all-mode map is absent.

4. **Objection: the factor in the energy current is wrong. DISMISSED.** Half-bond assignment and onsite-force cancellation give exactly $c/(4\chi)$ times the anticommutator; two independent derivations reproduce it.

5. **Objection: the current constant is merely a loose estimate. DISMISSED after sharpening.** Completing the simultaneous kinetic–bond square gives the sharp form constant $\sqrt{c/(2\chi)}$, a factor two below the first safe estimate.

6. **Objection: first-moment propagation proves common alpha. UPHELD as false.** Higher moments and thermodynamic Cauchy convergence remain open.

7. **Objection: divergent cutoff moments prove dynamics does not exist. UPHELD as false.** They refute only the named unweighted theorem/core routes.

8. **Objection: cubic polynomial growth alone proves an operator-norm obstruction. UPHELD as incomplete in the first draft and repaired.** Nonreal $z$, Weyl-displaced compact-support test vectors, fixed input norms and the expanding-ball support condition now give the explicit $R^3$ norm lower bound.

9. **Objection: this advances C6 or completes Pre-A. UPHELD as false.** Both result families are T0 and claim-nonbearing.

## 12 Reproduction

Round-1 primary, independent, and integrated audits:

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_round1_frozen_quadratic_causal_admission_triage.py

E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_round1_frozen_quadratic_causal_admission_triage_
independent.py

E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_round1_frozen_quadratic_causal_admission_triage_
verify.py
```

Weighted-energy primary, independent, and integrated audits:

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_common_local_derivation_weighted_
energy_route_split.py

E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_common_local_derivation_weighted_
energy_route_split_independent.py

E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_common_local_derivation_weighted_
energy_route_split_verify.py
```

The Round-1 routes are symbolic and independent exact-rational implementations. The weighted-energy routes independently reconstruct the Q3 graph, source envelope, local current, sharp form constant, cutoff lower bound, homogeneity obstruction and every open-scope firewall.

## 13 Result footer

```
Result ID:                PRE-A-ROUND1-T053-FROZEN-INVENTORY-AND-NO-
                          CURRENT-ADMITTED-MICROSCOPIC-SURVIVOR;
                          PA-CP1-ST8-Q3LOCK-COMMON-LOCAL-DERIVATION-
```

| | |
|---|---|
| | SOURCE-UNIFORM-WEIGHTED-FIRST-ENERGY-CONE-AND-FOURIER-CUTOFF-ROUTE-SPLIT. |
| Precise statement: | Bounded Round-1 inventory frozen, no current admitted survivor, no prospective validation; common local derivation and first weighted-energy cone proved; two unweighted routes rejected. |
| Scope: | T0 fixed-lattice and categorical route checkpoint. |
| Dependencies: | EXP-000780--EXP-000790; EXP-000791--EXP-000792. |
| Evidence grade: | ANALYTIC + EXACT EXECUTED + INDEPENDENT + PDF QA. |
| Reproduction command: | Run the six commands under Reproduction in the listed order, with each integrated verifier last. |
| Expected output: | Round-1 PASS 56/56, 95/95, 14/14; weighted-energy PASS 35/35, 111/111, 16/16. |
| Falsification gate: | Any hash, matrix, survivor, Q3 graph, coefficient, form square, cutoff growth, or scope mismatch. |
| Tier before / after: | C6 T1 / C6 T1; no claim or tier change. |
| No-overclaim statement: | Prospective validation, common alpha/KMS, algebraic grounds, GNS gap, continuum, physical empty, C6, CP1, Sector A and Pre-A remain open. |
| Next required action: | Freeze a future validation relation; prove higher weighted moments and thermodynamic Cauchy closure. |